

Analyzing the retraining frequency of global forecasting models: towards more stable forecasting systems.

Marco Zanotti^{a,*}

^aDepartment of Economics, Management and Statistics, University of Milano-Bicocca, Milan, Italy

Abstract

Forecasting systems based on global models are increasingly adopted in large-scale applications due to their strong predictive performance and scalability. However, in operational settings, forecasts are routinely updated through frequent model retraining, often without a clear understanding of its implications for forecast stability, accuracy, and computational cost. This study provides a large-scale empirical analysis of how retraining affects both point and probabilistic forecasts produced by global forecasting models. Using multiple state-of-the-art machine learning and deep learning approaches across three widely used forecasting datasets, we examine whether frequent retraining is necessary to maintain forecast quality. Our results show that reducing the retraining frequency rarely harms forecast accuracy and stability over time. In contrast, frequent retraining is often neither optimal nor required, especially for point forecasting, and only marginally beneficial for probabilistic forecasts. Moreover, the study shows that accuracy and stability are not necessarily conflicting objectives when adopting a global modeling approach. Overall, the findings suggest that global forecasting models are inherently robust to less frequent retraining, and that carefully chosen periodic update strategies can support the development of more stable and sustainable forecasting systems.

Keywords: Time series, Forecast stability, Vertical stability

*Corresponding author

Email address: zanottimarco17@gmail.com (Marco Zanotti)